5 **(a)** A particle of mass *m*, carrying a negative charge - q and travelling at speed *v*, enters a region of uniform magnetic field of flux density *B* directed at right angles to the motion of the particle as shown in Fig. 5.1.

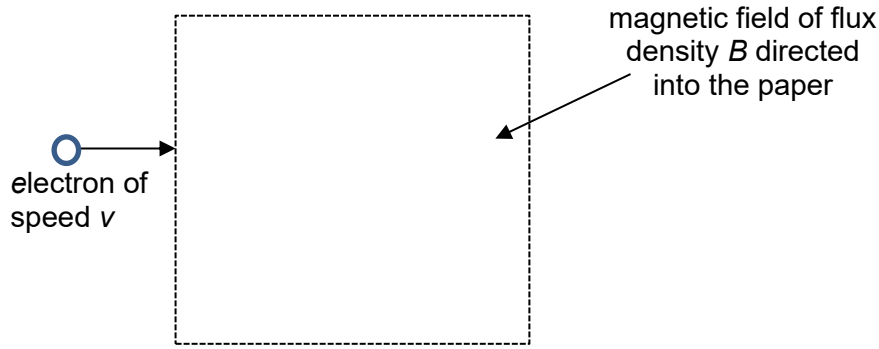

Fig. 5.1

(i) State the expression for the magnitude of the force *F* acting on the particle and the direction of the force.

F = ...

direction =..... [2]

(ii) Explain why the path of the electron is circular.

.....

.....

.....

..... [3]

(b) The diagram in Fig. 5.2 shows a type of cathode ray tube containing a small quantity of gas. Electrons from a hot cathode emerge from a small hole in the conical shaped anode, and the path subsequently followed is made visible by the gas in the tube.

The accelerating voltage is 5.0 kV.

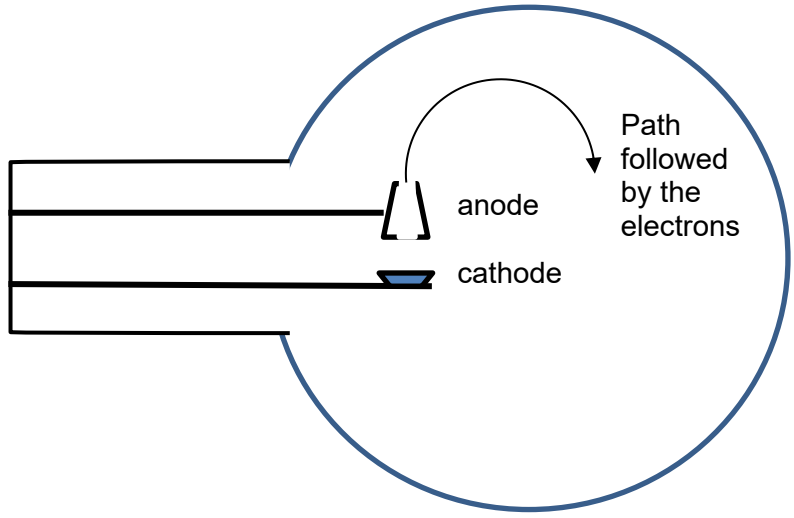

Fig. 5.2

(i) Calculate the speed of the electrons as they emerge from the anode.

speed = ……..m s^{-1} [2]

(ii) The apparatus is situated in a uniform magnetic field acting into the plane of the paper.

1. Calculate the radius of the circular path for a flux density of 2.0×10^{-3} T.

radius = …… m [2]

2. Suggest how the gas in the tube might make the path of the electrons visible.

……

…… [1]